

Micronutrients intake is associated with improved sperm DNA quality in older men.



OBJECTIVE: To investigate whether lifestyle factors such as increased dietary intake of micronutrients reduce the risks of sperm DNA damage, and whether older men benefit more than younger men. **DESIGN:** Cross-sectional study design with equalized assignments into age groups. **SETTING:** National laboratory and university. **PATIENT(S):** Nonclinical group of 22-80-year-old nonsmoking men (n = 80) who reported no fertility problems. **MAIN OUTCOME MEASURE(S):** Sperm DNA damage measured by alkaline and neutral DNA electrophoresis (i.e., sperm Comet assay). **RESULT(S):** Sociodemographics, occupational exposures, medical and reproductive histories, and lifestyle habits were determined by questionnaire. The average daily dietary and supplement intake of micronutrients (vitamin C, vitamin E, β -carotene, zinc, and folate) was determined using the 100-item Modified Block Food Frequency Questionnaire (FFQ). Men with the highest intake of vitamin C had approximately 16% less sperm DNA damage (alkaline sperm Comet) than men with the lowest intake, with similar findings for vitamin E, folate, and zinc (but not β -carotene). Older men (>44 years) with the highest vitamin C intake had approximately 20% less sperm DNA damage compared with older men with the lowest intake, with similar findings for vitamin E and zinc. The older men with the highest intake of these micronutrients showed levels of sperm damage that were similar to those of the younger men. However, younger men (<44 years) did not benefit from higher intakes of the micronutrients surveyed. **CONCLUSION(S):** Men with higher dietary and supplement intake of certain micronutrients may produce sperm with less DNA damage, especially among older men. This raises the broader question of how lifestyle factors, including higher intakes of antioxidants and micronutrients, might protect somatic as well as germ cells against age-associated genomic damage. Copyright © 2012. Published by Elsevier Inc.